CS491 – Fall 2027 – Project Information Form

Project information submitted for the CS491 Design project.

Email

ceylin.atalar@ug.bilkent.edu.tr

Name of the Project

GENIE5 (GEne Network Inference from Evidence)

One paragraph description of the project

Genie5 turns a biomedical research question into a directed, evidence-backed map of gene regulation. Statistical analysis can reveal that two genes are linked, but rarely which one regulates the other, so researchers still settle the question by hand-searching the literature, one gene pair at a time. Genie5 automates that review. It collects evidence from scientific papers and curated biological databases, reconciles conflicting reports, and determines the direction and type of each interaction, such as activation or repression. Researchers receive an interactive 2D/3D network that they can explore, verify and export to their existing tools, including CRNA for analysis on their own expression data.

Supervisor

Prof. Can Alkan, Department of Computer Engineering, Bilkent University

Innovation Expert

Asst. Prof. Gurkan Bebek, Case Center for Proteomics and Bioinformatics, Case Western Reserve University

Project Web Page

https://twin-pathways.github.io/

Team Members

Student ID	Name
22302668	Sırrı Kaan Oğuzkan
22302229	Derin Ceylin Atalar
22302289	Ayça Topkaya
22303934	Alper Koçyiğit
22302218	Ömer Turab Memiş